Chapter 1: Introduction to Data Structures

Technical / interview questions, exercises, multiple choice questions and answer key for Chapter 1 (Introduction to Data Structures) of Data Structures.

Technical / Interview Questions

1. Explain the concept of data and its importance in computer science.
2. What is data organization? Explain different ways of organizing data.
3. Define a data structure. Why are data structures required?
4. Explain the need for data structures in software development.
5. Describe the basic operations on data structures with examples.
6. Explain basic terminology used in data structures such as element, node, field, and record.
7. Discuss the advanced terminology related to data structures.
8. Classify data structures and explain each category with examples.
9. Differentiate between primitive and non-primitive data structures.
10. Compare linear and non-Linear Data Structures with suitable examples.
11. Explain the characteristics of Linear Data Structures.
12. Explain the characteristics of non-Linear Data Structures.
13. What factors should be considered while selecting a data structure?
14. Explain the significance of data structures in computer science.
15. Discuss real-world applications of data structures in areas such as oper ating systems, databases, and networking.
16. Explain the difference between static and dynamic data structures with suitable examples.
17. What is meant by time complexity and space complexity? Why are they important in data structure selection?
18. Explain Abstract Data Types (ADT) with an example. How is an ADT different from its implementation?
19. Compare arrays and linked lists in terms of memory allocation, access time, and applications.
20. Explain the role of algorithms in data structures. How are algorithms and data structures related?
21. Why are non-Linear Data Structures preferred for representing hierarchical and network-based data?
22. What are the advantages of using ADT-based design in programming?
23. Explain FIFO and LIFO principles with real-life examples.
24. How does data organization affect program efficiency and performance?
25. Give an example where choosing the wrong data structure can degrade system performance.

Exercises

1. Student Marks Management: A college needs to store and manage the marks of students in a class. Identify a suitable data structure for storing the marks and perform operations such as insertion of new marks, updating existing marks, and displaying all marks. Draw a neat diagram representing the chosen data structure.

2. Browser Navigation System: A web browser maintains the history of visited web pages to support back and forward navigation. Identify the appropriate data structure used for this purpose and explain how navigation operations are performed. Illustrate the working with a simple diagram.
3. Call Center Queue: A call center handles incoming customer calls in the order in which they arrive. Identify the data structure used to manage this system, name the operations involved in adding and removing calls, and trace the operations when multiple calls are received and one call is answered.
4. File System Organization: An operating system organizes files and folders in a hierarchical structure. Identify the suitable data structure for representing this organization, draw a simple diagram, and explain why a linear data structure is not appropriate for this scenario.
5. Undo Operation in a Text Editor: A text editor provides an undo feature to revert the most recent user actions. Identify the data structure used to implement this feature, explain how undo operations are carried out, and illustrate the process using a diagram.
6. City Road Network: A city road map contains multiple locations connected by roads. Identify the most suitable data structure to represent this system and explain how locations and connections are modeled. Mention real-life operations that can be performed using this structure.
7. Dynamic Data Storage: A program needs to store data whose size is not fixed and may increase or decrease during execution. Identify whether a static or dynamic data structure should be used, select an appropriate data structure, and justify why arrays are not suitable in this case.
8. Online Shopping Cart: An online shopping application maintains a list of items added by a user to the cart. Identify a suitable data structure and explain how items are added, removed, and displayed. Justify your choice of data structure.
9. Library Book Organization: A library organizes books based on categories and subcategories. Identify the data structure used for this purpose, draw a simple representation, and explain how this structure supports efficient searching.
10. Data Structure Selection Challenge: Design a system that requires fast access, frequent insertion and deletion, and dynamic size. Identify the most suitable data structure, list its advantages, and mention one limitation.
11. Train Reservation System: A railway reservation system needs to manage passenger requests dynamically. Identify suitable data structures for booking, cancellation, and waiting list management. Justify your choices.
12. Expression Evaluation: Given an arithmetic expression, identify the data structure used to evaluate the expression and explain the evaluation process step by step.
13. Social Media Friend Network: A social media platform maintains user profiles and connections between users. Identify an appropriate data structure to represent this system and explain its advantages.
14. Memory Allocation Simulation: Simulate a simple memory manager using a suitable data structure. Explain how memory blocks are allocated and deallocated.
15. Menu-Driven Program Design: Design a menu-driven program that allows insertion, deletion, traversal, and searching of elements using any linear data structure of your choice.

Multiple Choice Questions

1. What is meant by data?
    A. Processed information
    B. Raw facts and figures
    C. Computer programs
    D. Instructions to hardware
2. Data organization refers to:
    A. Storing data randomly
    B. Arranging data for efficient processing
    C. Encrypting data
    D. Deleting unnecessary data
3. Which of the following best defines a data structure?
    A. A programming language
    B. A collection of data and operations on it
    C. A compiler
    D. A database
4. What is a linear data structure?
    A. A structure that represents hierarchical relationships
    B. A structure where elements are arranged in a linear sequence
    C. A structure that supports only numerical data
    D. A structure with random data access
5. Which of the following is an example of a linear data structure?
    A. Binary Tree
    B. Graph
    C. Stack
    D. Hash Table
6. Which of the following is NOT a linear data structure?
    A. Array
    B. Stack
    C. Queue
    D. Tree
7. Which data structure follows the LIFO principle?
    A. Queue
    B. Stack
    C. Array
    D. Tree
8. Which data structure follows the FIFO principle?
    A. Stack
    B. Queue
    C. Tree
    D. Graph
9. Which of the following is a non-linear data structure?
    A. Array
    B. Linked List
    C. Stack
    D. Tree

10. Which data structure is best suited for hierarchical data representation?

A. Array
B. Stack
C. Queue
D. Tree

11. Which of the following is a primitive data structure?

A. Array
B. Stack
C. Integer
D. Linked List

12. Which of the following is a non-primitive data structure?

A. Integer
B. Float
C. Character
D. Array

13. Which operation is used to access all elements of a data structure?

A. Insertion
B. Deletion
C. Traversal
D. Sorting

14. Which operation is used to add an element to a data structure?

A. Traversal
B. Insertion
C. Deletion
D. Searching

15. Which operation removes an element from a data structure?

A. Traversal
B. Sorting
C. Deletion
D. Merging

16. Which operation arranges elements in a specific order?

A. Searching
B. Traversal
C. Sorting
D. Insertion

17. Which operation combines two data structures into one?

A. Sorting
B. Traversal
C. Deletion
D. Merging

18. Which data structure uses contiguous memory allocation?

A. Linked List
B. Tree
C. Graph
D. Array

19. Which data structure uses pointers to connect elements?

A. Array
B. Stack
C. Linked List
D. Queue

20. Which of the following is a dynamic data structure?

A. Static array
B. Array
C. Linked List
D. Fixed-size stack

21. What does ADT stand for?

A. Advanced Data Types
B. Abstract Data Types
C. Algorithmic Data Types
D. Array Data Types

22. Which of the following is NOT an ADT?

A. Stack
B. Queue
C. Integer
D. List

23. What is the purpose of time complexity analysis?

A. To measure program execution speed
B. To analyze algorithm efficiency with respect to input size
C. To compare programming languages
D. To calculate memory size

24. Which notation is used to express the upper bound of algorithm complexity?

A. Ω (Omega)
B. Θ (Theta)
C. O (Big O)
D. p (Pi)

25. What does O(1) time complexity represent?

A. Linear time
B. Constant time
C. Quadratic time
D. Exponential time

26. In terms of time complexity, which is most efficient?

A. O(n)
B. O(log n)
C. $O(n^2)$
D. $O(2^n)$

27. Space complexity analysis mainly focuses on:

A. Execution time
B. Storage required with respect to input size
C. Compiler efficiency
D. CPU speed

28. Which of the following is NOT considered in space complexity?

A. Variables
B. Constants
C. Input size
D. Execution time

29. Which factor is most important while selecting a data structure?

A. Programming language
B. Time and memory efficiency
C. Compiler type
D. Hardware brand

30. Which of the following is a real-world application of data structures?

A. File systems
B. Artificial intelligence
C. Social networks
D. All of the above

31. Which data structure allows insertion and deletion at both ends?

A. Stack
B. Queue
C. Deque
D. Array

32. Which of the following operations is not possible in constant time in a singly linked list?

A. Insertion at beginning
B. Deletion at beginning
C. Accessing an element by index
D. Traversal

33. Which data structure is best suited for undo–redo operations?

A. Queue B. Array
C. Stack
D. Tree

34. Which of the following is an advantage of linked lists over arrays?

A. Faster random access
B. Contiguous memory allocation
C. Dynamic memory allocation
D. Less memory usage

35. Which data structure is commonly used to implement recursion?

A. Queue
B. Stack
C. Tree
D. Graph

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	2	B	3	B	4	B	5	C
6	D	7	B	8	B	9	D	10	D
11	C	12	D	13	C	14	B	15	C
16	C	17	D	18	D	19	C	20	C

21	B	22	C	23	B	24	C	25	B
26	B	27	B	28	D	29	B	30	D
31	C	32	C	33	C	34	C	35	B